UNIT 9 · TOPIC 4

Data Acquisition, Storage and Dynamic Measurements

Syllabus topic: Data acquisition and storage; dynamic measurements

Data acquisition (DAQ) and storage

An ANALOG-TO-DIGITAL CONVERTER (ADC) samples the conditioned signal. The DAQ chain:

1. Sensor.
2. Signal conditioning.
3. Multiplexer.
4. Sample-and-hold.
5. ADC.
6. Computer — logs, displays, stores and analyses (data may also be telemetered).

NYQUIST–SHANNON theorem: sampling rate $f_s > 2 \cdot f_{max}$

- Sampling slower than twice the highest signal frequency causes **ALIASING** — a false, lower-frequency signal. Hence an anti-alias filter BEFORE the ADC.
- **ADC resolution:** n bits give 2^n levels — a 12-bit ADC gives **4096** steps.

Dynamic measurements

Time-varying quantities — vibration, fluctuating draft, pressure pulses — need transducers with adequate FREQUENCY RESPONSE:

- The system's natural frequency must lie well ABOVE the highest signal frequency.
- Damping of ≈ **0.6–0.7** of critical gives a flat response without overshoot.
- Use fast sampling and proper anti-alias filtering.
- **Typical dynamic sensors:** piezoelectric accelerometers (vibration), pressure transducers, and strain-gauge dynamometers with a high natural frequency.

Practice MCQs

Q27. An ADC converts:

(a) Analog signal to digital
(b) Digital to analog
(c) AC to DC
(d) Force to voltage

Q28. The Nyquist criterion requires the sampling rate to be:

(a) Equal to signal frequency
(b) Greater than twice the highest signal frequency
(c) Half the frequency
(d) Any value

Q29. Sampling too slowly causes:

(a) Amplification
(b) Aliasing (false low-frequency signal)
(c) Filtering
(d) Higher accuracy

Q30. An anti-aliasing filter is placed:

(a) After the ADC
(b) Before sampling/ADC
(c) In the display
(d) Nowhere

Q31. A 12-bit ADC provides how many quantisation levels?

(a) 12
(b) 144
(c) 1024
(d) 4096

Q32. For dynamic (time-varying) measurement, the sensor's natural frequency should be:

(a) Below the signal frequency
(b) Well above the highest signal frequency
(c) Equal to it
(d) Zero

Q33. A damping ratio giving a flat, fast response without overshoot is about:

(a) 0
(b) 0.1
(c) 0.6–0.7
(d) 2.0

Q34. A piezoelectric accelerometer is typically used to measure:

(a) Temperature
(b) Vibration/acceleration
(c) Flow
(d) pH

Answers

Q27 — (a) Analog signal to digital

Q28 — (b) Greater than twice the highest signal frequency — *Nyquist: $f_s > 2f_{max}$.*

Q29 — (b) Aliasing (false low-frequency signal) — *Too slow → aliasing.*

Q30 — (b) Before sampling/ADC — *Anti-alias filter before the ADC.*

Q31 — (d) 4096 — *12-bit = 2^{12} = 4096 levels.*

Q32 — (b) Well above the highest signal frequency — *Dynamic sensing needs natural frequency ≫ signal frequency.*

Q33 — (c) 0.6–0.7 — *Damping ~0.6–0.7 for a flat response.*

Q34 — (b) Vibration/acceleration

Sources: Doebelin — Measurement Systems; Beckwith — Mechanical Measurements; Sawhney — Electrical & Electronic Measurements; HBM/Omega strain-gauge technical notes (as given in the source notes, July 2026); ASAE instrumentation standards; NPTEL measurement lectures.